Development of a Bayesian Optimization algorithm to explore magnetic field topology for radiation dose reduction in crewed space mission shielding.

Introduction

The problem of radiation shielding is of paramount importance in crewed space missions: the limitation of damages to the astronauts' healths puts limits on the duration of a crewed mission. At this time, NASA duration for ISS missions is 6 months [1]. This cap makes unfeasible, at the current stage, the possibility of planning crewed mission to explore other Solar System's planets.

There are two types of radiation shielding: passive and active. Passive shielding operates by placing some shielding material that absorbs radiation by particle interaction. This is the type of shielding that the most research has been conducted about.

Active radiation shielding relies on a magnetic field to deflect incoming particles via Lorentz force. Some designs have been proposed, like in [2–4]. Also, preliminary studies have been conducted by NASA concluding that using magnetic fields generated by high temperature superconducting coils is possible, while the proposed designs do not offer significant improvements over passive shielding [5]. At the current state, no algorithms for BO in magnetic field optimization have been proposed in the reviewed literature.

In the current literature, the most widely used method for magnetic field optimization is topology optimization, with applications to finding the best coil distribution for the largest uniform field in [6] and in reverse magnetostatics to find optimal configuration of coils to generate a target field in [7]. This may prove useful to reverse engineer an optimal solution from Bayesian Optimization. Given the experience of the RMIT Space Physics Group in this challenge, as in [8–10], this research proposes using a statistical optimization model to explore exotic field topologies for minimizing the received radiation dose in Monte Carlo simulations. Due to the complex and non-linear nature of dependencies between field topology and shielding capabilities, no magnetic field optimization has been applied to this task. However we feel that the huge exploration capabilities, even when searches are expensive, and the ability to operate in a black box scenario of BO may provide an important insight in designing the shield configuration.

Research Questions:

1. Is there an optimal field topology that minimizes the probability of particle penetration and can it be found via Bayesian Optimization?
2. What low-dimensional parametrizations of magnetic fields produces a physically meaningful optimum while enabling efficient Bayesian Optimization?
3. What theoretical guarantees can be established for Gaussian Process–based Bayesian Optimization in this PDE-constrained electromagnetic design problem, where the objective is noisy, non-smooth, and induced by stochastic charged-particle transport?

Methods

The main mathematical tool is the Gaussian Process–based Bayesian Optimization, which tries to minimize the functional $\mathcal{J}(\theta)$ by observing data from it and without having to know it [11]. Since the simulation of $\mathcal{J}$ is complex, this avoids non-linearity and non-differentiability issues of classical optimization algorithms. The algorithm works as follow:

Algorithm 1 Basic BO minimization

```
Place a Gaussian process prior on J.
Observe J at an n_0 number of points and set n = n_0.
while (n ≤ N) do
    Update posterior probability distribution of J, using all available data.
    Let θ_n be a maximizer of an acquisition function over θ.
    Observe y_n = J(θ_n).
    Increment n.
end while
Return point with highest evaluated J(θ)
```

The first task will be to find a parametrization of the **B** field that is both mathematically tractable and physically informative, given that BO works better in low dimension (max 50 params. approx.).

Then the algorithm will be implemented in a python code that will pass the field parameters to a C++ Monte Carlo simulations (the $\mathcal{J}$ functional), that will return the received radiation dose. The RMIT Space Physics group has provided the computational resources.

The RMIT Space Physics group has already developed simulators for this task that run on the particle physics software GEANT4, that can be used for the task.

A theoretical study on convergence guarantees and robustness under noise will be conducted in parallel, with the help of italian supervisors [12].

Contents

1 The Bayesian framework: preliminary statistics concepts

2 Gaussian Processes theory

2.1 Basic definitions

Definition 1 (Random Process)**.** *A random process, or stochastic process, is a collection of random variables $(X_t)_{t\in T}$ on the same probability space, and indexed by elements t of a set T. If the set T is a continuous interval, then the process is time-continuous.*

Definition 2 (Gaussian Process)**.** *A continuous time random process $(X_t)_{t\in T}$ is called a Gaussian process if for every finite subset $T_0 \subset T$ (i.e.: for any finite collection of indexes $\{t_1, ..., t_n\}$), the random vector:*

$$(X_t)_{t\in T_0}$$

has normal distribution. We indicate the process as:

$$X(t) \sim \mathcal{GP}(m(\cdot), k(\cdot, \cdot)),$$

where $m(\cdot)$ is the mean function and $k(\cdot, \cdot)$ is the covariance function.

Observation 1. *A Gaussian process is completely specified by its mean and covariance function [13].*

We consider the Gaussian process as a distribution over the function space (the functions are induced by data), so we will write:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')),\ \forall x \in \mathcal{X}$$

where:

$$m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})],$$
$$k(\mathbf{x}, \mathbf{x}') = \mathbb{E}[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{x}') - m(\mathbf{x}'))].$$

Here the random variable $f(\mathbf{x})$ is the value of the function at point $\mathbf{x}$. In our case, the index space $\mathcal{X}$ is not a time interval, but a subset of $\mathbb{R}^D$, since our data are spatial.

2.2 Inference with GPs

The kernel covariance function k, implies the distribution over functions of the process. Assume that we have sample functions from the GP at any number of point. We can choose $X_\star$ as the set of $n_\star$ input points $\mathbf{x}_\star$. If the covariance function k is specified, we can write the covariance matrix of the set $X_\star$ and generate a random Gaussian vector.

$$\mathbf{f}_\star \sim \mathcal{N}(m(X_\star), K(X_\star, X_\star)). \tag{1}$$

After observing $\mathbf{f}_\star$, either directly or through random noise, we can condition the Gaussian Process in order to update our beliefs, following the principles of bayesian statistics.

2.2.1 Noise-free observations

If the observations are noise free we get the *joint prior distribution* of $\mathbf{f}, \mathbf{f}_\star$ according to the prior as:

$$\begin{pmatrix} \mathbf{f} \\ \mathbf{f}_\star \end{pmatrix} \sim \mathcal{N}\left(\begin{matrix} m(X) \\ m(X_\star) \end{matrix}, \begin{bmatrix} K(X,X) & K(X,X_\star) \\ K(X_\star,X) & K(X_\star,X_\star) \end{bmatrix} \right), \tag{2}$$

where $K(X, X_\star)$ is the $n \times n_\star$ matrix of the covariances evaluated at all pairs of n training points and $n_\star$ test points. The analogue reasoning is valid for $K(X, X)$, $K(X_\star, X)$ and $K(X_\star, X_\star)$.

After observing the points, we can get the *posterior distribution* by restricting the prior to the functions that agree with observed points. Here the assumption that the evaluations are noise-free becomes critical. We can condition the Gaussian prior on the observations, following bayesian theory, to obtain:

$$\mathbf{f}_\star | X_\star, X, \mathbf{f} \sim \mathcal{N}(m(X_\star) + K(X_\star, X)K(X, X)^{-1}\mathbf{f},\ K(X_\star, X_\star) - K(X_\star, X)K(X, X)^{-1}K(X, X_\star))\ . \tag{3}$$

2.2.2 Noisy observations

In many real world applications, we must account for some random noise on the observed data, hence $y = f(\mathbf{x}) + \varepsilon$. We will assume that ε is additive i.i.d. Gaussian noise with variance σ_n^2. The covariance is given by:

$$\operatorname{cov}(\mathbf{y}) = K(X, X) + \sigma_n^2 I, \tag{4}$$

We can rewrite equation (2) as:

$$\begin{pmatrix} \mathbf{y} \\ \mathbf{f}_\star \end{pmatrix} \sim \mathcal{N}\left(\begin{matrix} m(X) \\ m(X_\star) \end{matrix} , \begin{bmatrix} K(X,X) + \sigma_n^2 I & K(X, X_\star) \\ K(X_\star, X) & K(X_\star, X_\star) \end{bmatrix} \right). \tag{5}$$

We now want to get a posterior distribution, as in (3):

$$\begin{aligned} \mathbf{f}_\star | X_\star, X, \mathbf{y} \sim \mathcal{N}(&m(X_\star) + K(X_\star, X)[K(X,X) + \sigma_n^2 I]^{-1}\mathbf{y}, \\ &K(X_\star, X_\star) - K(X_\star, X)[K(X,X) + \sigma_n^2 I]^{-1} K(X, X_\star)) \ . \end{aligned} \tag{6}$$

We can use a more compact notation by writing $K = K(X, X)$ and $K_\star = K(X, X_\star)$. Furthermore, if there is only one test point $\mathbf{x}_\star$, as will be the case in Bayesian Optimization, the covariance matrix $K_\star$ is reduced to a vector of covariances between $\mathbf{x}_\star$ and the n training points. We will call this vector $\mathbf{k}_\star$. Then:

$$\begin{aligned} \mathbb{E}[\mathbf{f}_\star | \mathbf{x}_\star, X, \mathbf{y}] &= \mathbf{k}_\star^\top (K + \sigma_n^2 I)^{-1} \mathbf{y}, \\ \operatorname{var}(\mathbf{f}_\star | \mathbf{x}_\star, X, \mathbf{y}) &= k(\mathbf{x}_\star, \mathbf{x}_\star - \mathbf{k}_\star^\top (K + \sigma_n^2 I)^{-1} \mathbf{k}_\star. \end{aligned} \tag{7}$$

2.3 Kernel functions

2.3.1 Definition and requirements

While in most of applications the mean function is taken to be uniformly 0, the covariance function needs to be well chosen because, as we've seen from the prediction section, it encodes our beliefs about the function that we're learning. The covariance function, defines the notion of nearness or similarity for GPs by providing a measure of how much two points are correlated according to their distance. In general, an arbitrary function of $\mathbf{x}, \mathbf{x}'$ will not be a valid covariance function, but it will be subject to some requirements.

In general, a function that maps a pair of inputs $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$ to $\mathbb{R}$ is called a *kernel*. The name comes from integral operators theory, where the operator:

$$(T_k f)(\mathbf{x}) = \int_{\mathcal{X}} k(\mathbf{x}, \mathbf{x}') f(\mathbf{x}') \ d\mu(\mathbf{x}'). \tag{8}$$

A real kernel is *symmetric* if $k(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}', \mathbf{x})$. Clearly, covariance functions must be symmetric by definition.

A kernel is also *positive semi-definite* if:

$$\int_{\mathcal{X}} k(\mathbf{x}, \mathbf{x}') f(\mathbf{x}) f(\mathbf{x}') \ d\mu(\mathbf{x}) \ d\mu(\mathbf{x}') \geq 0 \ \forall f \in L_2(\mathcal{X}, \mu). \tag{9}$$

Valid covariance functions must be positive semi-definite.

Also, a kernel is *stationary* if it's a function of $\mathbf{x} - \mathbf{x}'$, thus it's invariant to translations in the input space. Furthermore, if it's a function of $|\mathbf{x} - \mathbf{x}'|$, then it's also *isotropic*, which means it's invariant to all rigid motions.

Another property that will be extremely useful in analyzing kernels is the following.

Definition 3 (Mean square continuity)**.** *Let $\mathbf{x}_1, \mathbf{x}_2, ...$ be a sequence of points and $\mathbf{x}_\star \in \mathbb{R}^D$ a fixed point such that $|\mathbf{x}_k - \mathbf{x}_\star| \to 0$ as $k \to \infty$. Then a process $f(\mathbf{x})$ is said to be* ***mean square continuous*** *at $\mathbf{x}_\star$ if $\mathbb{E}[|f(\mathbf{x}_k) - f(\mathbf{x}_\star)|^2] \to 0$ as $k \to \infty$.*

Proposition 1. *A random field is mean square continuous in $\mathbf{x}_\star$ if and only if its kernel $k(\mathbf{x}, \mathbf{x}')$ is continuous at $\mathbf{x} = \mathbf{x}' = \mathbf{x}_\star$. For a stationary kernel, this reduces to check continuity at $\mathbf{x} = 0$.*

Definition 4 (Mean square differentiability)**.** *The mean square derivative of $f(\mathbf{x})$ in the ith direction is defined as:*

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = \lim_{h \to 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - \mathbf{x}}{h},$$

when the limit exists.

We now present a couple of kernel classes that will be of interest throughout this work. For simplicity's sake, in the next sections, we will pose $r = \mathbf{x} - \mathbf{x}'$

2.3.2 The γ-exponential kernel class

The γ-exponential class is given by the expression:

$$k(r) = exp(-(r/\ell)^\gamma), \quad \text{for } 0 < \gamma \leq 2. \tag{10}$$

For $\gamma = 2$, this gives the *Squared Exponential* kernel, one of the most used in literature:

$$k_{SE}(r) = exp\left(-\frac{r^\gamma}{2\ell^\gamma}\right), \tag{11}$$

where ℓ is the characteristic length scale. This kernel is infinitely differentiable, so the GP with this covariance function has mean squared derivatives of all orders, implying that it's very smooth. This usually mean that this specific kernel is not suitable for modelling physical real world objective functions that are rarely infinitely smooth.

2.3.3 The Matérn kernel class

The Matérn kernel class has the following expression.

$$k_M(r) = \frac{2^{(1-\nu)}}{\Gamma(\nu)}\left(\frac{\sqrt{2\nu}r}{\ell}\right)^\nu K_\nu\left(\frac{\sqrt{2\nu}r}{\ell}\right), \tag{12}$$

with positive parameters ν and l. K_ν is a modified Bessel function. In this class, the process is k-times Mean Square differentiable with $k < \nu$. When $\nu = n + \frac{1}{2}$, $n \in \mathbb{N}$ the Bessel function expression becomes simple to evaluate. In particular, the most widely used Matern kernels are for $\nu = 5/2$ and $\nu = 3/2$, with expressions:

$$k_{3/2}(r) = \left(1 + \frac{\sqrt{3}r}{\ell}\right)\exp\left(-\frac{\sqrt{3}r}{\ell}\right), \tag{13}$$

$$k_{5/2}(r) = \left(1 + \frac{\sqrt{5}r}{\ell} + \frac{5r^2}{3l^2}\right)\exp\left(-\frac{\sqrt{5}r}{\ell}\right). \tag{14}$$

The first one is used for rough physical processes, while the latter is suitable for more smooth process and more widely used. It's also the default choice for most Matern class kernel implementation in GPs libraries for Python and R.

2.4 Regression with Gaussian Processes

3 Bayesian Optimization theory

4 Preliminary study on objective function continuity

As the literature highlights, the only requirement for Bayesian Optimization is for the objective function to be continuous, otherwise the algorithm may not converge [11]. We suppose now that a parametrization such that $\mathbf{B}(\theta)$ is continuous w.r.t. $\theta \in \Theta$ exists. This is actually obvious, since we can always construct our parametrization to be continuous, by ignoring the physical meaning. Formally, we can say that if $\mathcal{F}$ is the space of all vector fields $\mathbb{R}^3 \to \mathbb{R}^3$, then $\mathbf{B} \in C^0(\Theta, \mathcal{F})$.

Now the theoretical challenge is to make the radiation dose to depend continuously on the magnetic field. We will tackle this problem by dividing it in two pieces.

4.1 Trajectory continuity

The first point is to make sure that the trajectory of a generic charged particle depends continuously on the magnetic field. The trajectory is defined as the unique solution $\mathbf{x}(t; \mathbf{B}, \xi)$ to the initial value problem governed by the Lorentz force:

$$\begin{cases} \mathbf{x}(0) = \mathbf{x}_0, \ \dot{\mathbf{x}}(0) = \mathbf{v}_0, \\ m\frac{d^2\mathbf{x}}{dt^2} = q\left(\frac{d\mathbf{x}}{dt} \times \mathbf{B}_\theta(\mathbf{x})\right), \end{cases} \tag{15}$$

where $\mathbf{B}_\theta$ is a functional in the Banach space $C^0(\Theta, \mathcal{F})$ of mappings between the parameters space and the 3 dimensional vector fields [1]. For briefness, it will hereon be called $\mathbf{B}$. $\xi = (\mathbf{x}_0, \mathbf{v}_0) \in \Xi$ represents the initial state of a particle in the probability space of incoming radiation.

[1]Completeness is ensured by compactness of Θ, which is required by Bayesian Optimization.

Let's now rewrite the system as a first order ODE problem. Let $\gamma(t) = (\mathbf{x}(t), \mathbf{v}(t))$ be the trajectory of a particle in the field, then:

$$\begin{cases} \gamma(0) = \xi, \\ \frac{d\gamma}{dt} = F(\gamma, \mathbf{B}, t). \end{cases} \tag{16}$$

where:

$$F(\gamma, \mathbf{B}, t) = \begin{pmatrix} \mathbf{v}(t) \\ \frac{q}{m}\left(\mathbf{v}(t) \times \mathbf{B}(\mathbf{x})\right) \end{pmatrix} \tag{17}$$

is the linear functional representing Lorentz force.
We want to prove the following.

Proposition 2. *Let $\gamma(t)$ be a solution of problem* (16)*, then:*

1. *$\gamma(t)$ exists and is unique,*
2. *$\gamma(t)$ depends on $\theta \in \Theta$ with uniform continuity.*

Let's first introduce a technical lemma that will be used in¸ the proof:

Lemma 1 (Gronwall's inequality)**.** *Let I be an interval on the real line in the form $[a, b]$, $[a, \infty)$, $[a, b)$ with $a < b$. Let α, β, u be real valued functions defined on I. If these conditions are true:*

1. *u, β are continuous,*
2. *The negative part of α is integrable on every closed and bounded subinterval of I,*
3. *β is non-negative and α is non-decreasing,*
4. *u satisfies the integral inequality:*

$$u(t) \leq \alpha(t) + \int_a^t \beta(s)u(s)\ ds\ t \in I\ ,$$

then the following bound holds:

$$u(t) \leq \alpha(t) \exp\left(\int_a^t \beta(s)\ ds\right)\ . \tag{18}$$

Proof. We can exploit the following facts:

1. $\mathbf{B}$ is continuous and Lipschitz by construction;
2. $\|\mathbf{v}\|$ is physically bounded by the speed of light;

to apply the Picard-Lindelof theorem, which ensures existence and unicity of solution $\gamma(t)$.

Now that we know $\gamma(t)$ exists and is unique, we can proceed on proving that it also depends with uniform continuity on $F(\gamma, \mathbf{B}, t)$. Since the speed $\|\mathbf{v}\|$ is bounded by the speed of light c and the magnetic field $\mathbf{B}$ is Lipschitz continuous, the functional $F(\gamma, \mathbf{B}, t)$ is globally Lipschitz in γ uniformly with respect to t, with Lipschitz constant L. Furthermore, F is continuously dependent on the parameter $\mathbf{B}$.

Let $\gamma_1(t)$ and $\gamma_2(t)$ be solutions corresponding to two different initial states $\xi_1, \xi_2 \in \Xi$ and two different magnetic fields $\mathbf{B}_1, \mathbf{B}_2$. By the integral formulation of the initial value problem, we have:

$$\gamma_i(t) = \xi_i + \int_0^t F(\gamma_i(s), \mathbf{B}_i, s)\, ds, \quad \text{for } i \in \{1, 2\}. \tag{19}$$

Taking the norm of the difference and applying the triangle inequality yields:

$$\|\gamma_1(t) - \gamma_2(t)\| \leq \|\xi_1 - \xi_2\| + \int_0^t \|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_2(s), \mathbf{B}_2, s)\|\, ds. \tag{20}$$

By adding and subtracting $F(\gamma_1(s), \mathbf{B}_2, s)$ inside the integral:

$$\|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_2(s), \mathbf{B}_2, s)\| = \|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_1(s), \mathbf{B}_2, s) + F(\gamma_1(s), \mathbf{B}_2, s) - F(\gamma_2(s), \mathbf{B}_2, s)\| \leq$$
$$\|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_1(s), \mathbf{B}_2, s)\| + \|F(\gamma_1(s), \mathbf{B}_2, s) - F(\gamma_2(s), \mathbf{B}_2, s)\|\ .$$

Now we exploit the Lipschitz continuity of F with respect to γ and its linearity with respect to $\mathbf{B}$:

1. $\|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_1(s), \mathbf{B}_2, s)\| = \|F(\gamma_1(s), \mathbf{B}_1 - \mathbf{B}_2, s)\| \leq K\|\mathbf{B}_1(\mathbf{x}_1) - \mathbf{B}_2(\mathbf{x}_1)\|_\infty$,
2. $\|F(\gamma_1(s), \mathbf{B}_2, s) - F(\gamma_2(s), \mathbf{B}_2, s)\| \leq L\|\gamma_1(s) - \gamma_2(s)\|$,

where $K = \frac{qc}{m}$ is a constant derived from the physical bounds of the Lorentz force. This provides the bound:

$$\|F(\gamma_1(s), \mathbf{B}_1, s) - F(\gamma_2(s), \mathbf{B}_2, s)\| \leq L\|\gamma_1(s) - \gamma_2(s)\| + K\|\mathbf{B}_1(\mathbf{x}_1) - \mathbf{B}_2(\mathbf{x}_1)\|_\infty, \tag{21}$$

Substituting this back into the integral, we get:

$$\|\gamma_1(t) - \gamma_2(t)\| \leq \|\xi_1 - \xi_2\| + K\|\mathbf{B}_1 - \mathbf{B}_2\|_\infty t + L\int_0^t \|\gamma_1(s) - \gamma_2(s)\|\, ds. \tag{22}$$

We need now to apply Gronwall's lemma. Let:

1. $u(t) = \|\gamma_1(t) - \gamma_2(t)\|$,
2. $\alpha(t) = \|\xi_1 - \xi_2\| + K\|\mathbf{B}_1 - \mathbf{B}_2\|_\infty t$,
3. $\beta(t) = L$.

Then, condition 1 is clearly satisfied by continuity of the norm for u, while β is constant. Condition 2 holds because $\alpha(t) \geq 0\ \forall t \geq 0$, so the negative part of α on any interval $[0, T]$ is 0. We can also see that $\beta(t) = L > 0$ by definition of Lipschitz constant and $\alpha'(t) = K\|\mathbf{B}_1 - \mathbf{B}_2\|_\infty \geq 0$ by definition, thus infering condition 3. Condition 4 is satisfied by equation (22).

Applying Grönwall's inequality, we obtain the final bound:

$$\|\gamma_1(t) - \gamma_2(t)\| \leq (\|\xi_1 - \xi_2\| + K\|\mathbf{B}_1 - \mathbf{B}_2\|_\infty t)\, e^{Lt}. \tag{23}$$

This result shows that as $\xi_1 \to \xi_2$ and $\mathbf{B}_1 \to \mathbf{B}_2$, the difference between the trajectories $\|\gamma_1(t) - \gamma_2(t)\| \to 0$ uniformly on any compact time interval $[0, T]$. Thus, the trajectory depends with uniform continuity on both the initial conditions and the magnetic field.

Formally, we can define $\|\xi_1 - \xi_2\| < \delta_1$ and $\|\mathbf{B}_1 - \mathbf{B}_2\|_\infty < \delta_2$, with $\delta_1, \delta_2 > 0$, and:

$$\varepsilon = (\delta_1 + K\delta_2 t)e^{Lt},$$

hence:

$$\|\gamma_1(t) - \gamma_2(t)\| \leq (\delta_1 + K\delta_2 t)e^{Lt} = \varepsilon\ .$$

It is clear that for any choice of $\varepsilon > 0$ it is possible to find $\delta_1, \delta_2 > 0$ at any given time $t \in [0, T]$, thus concluding the continuity proof.

□

4.2 Revelator continuity

To ensure the continuity of the objective function, we need to avoid using a hit/miss detector, since step function is non continuous. We define a soft scoring kernel $\Phi : \mathbb{R}^3 \to [0, 1]$. A possible choice for ensuring C^∞ regularity is a Normalized Compact Mollifier with support on the ball $\mathcal{B}_\varepsilon$ of radius ϵ and center in the origin:

$$\Phi(\mathbf{x}) = \begin{cases} \frac{1}{\lambda_n(\mathcal{B}_\varepsilon)} \exp\left(-\frac{1}{1-\|\mathbf{x}/\epsilon\|^2}\right) & \text{if } \|\mathbf{x}\| < \epsilon, \\ 0 & \text{otherwise;} \end{cases} \tag{24}$$

where $\lambda_n(\mathcal{B}_\varepsilon) = \int_{\mathcal{B}_\varepsilon} \exp\left(-\frac{1}{1-\|\mathbf{x}/\epsilon\|^2}\right) d\lambda_n$, with λ_n as the n-dimensional Lebesgue measure, is the volume of the ball. Note: in this setup, the mollifier is also a probability distribution. This may be useful later.

The individual dose $D(\mathbf{B}, \xi)$ received for a single particle is then defined as the line integral along its trajectory $\gamma(s)$ times its kinetic energy:

$$D(\mathbf{B}, \xi) = \int_\gamma E(\xi)\Phi(\mathbf{x}(t; \mathbf{B}, \xi))ds = \int_0^T E(\xi)\Phi(\mathbf{x}(t; \mathbf{B}, \xi)) \cdot \|\mathbf{v}(t; \mathbf{B}, \xi)\|dt. \tag{25}$$

Since Boris integrator should conserve energy [14], we can eliminate time dependence from the speed norm, as it should be almost constant, thus obtaining:

$$D(\mathbf{B}, \xi) = E(\xi)\|\mathbf{v}(\xi)\| \int_0^T \Phi(\mathbf{x}(t; \mathbf{B}, \xi))\ dt. \tag{26}$$

This allows us to reduce calls to `Eigen::Vector3d.norm()`.

To conclude this discussion, the global objective function to be minimized is the expected received dose $\mathcal{J}(\boldsymbol{\Theta})$, defined by the expectation over the radiation spectrum $P(\xi)$:

$$\mathcal{J}(\boldsymbol{\Theta}) = \mathbb{E}_{\xi\sim P}[D(\mathbf{B},\xi)] = \int_{\Xi} D(\mathbf{B},\xi)P(\xi)\ d\xi = \int_{\Xi} E(\xi)\|\mathbf{v}(\xi)\|P(\xi)\int_0^T \Phi(\mathbf{x}(t;\mathbf{B},\xi))\ dt\ d\xi. \tag{27}$$

We take the expectation of the dose to try to average out the intrinsic variance of the Monte Carlo simulation. At this point, our observations will hopefully come from a continuous function (maybe even smooth) with some small random noise.

It is possible to transform the integral over Ξ to get a more explicit version:

$$\mathcal{J}(\boldsymbol{\Theta}) = \frac{1}{N}\sum_{\xi\in\Xi}\left(E(\xi)\|\mathbf{v}(\xi)\|\int_0^T \Phi(\mathbf{x}(t;\mathbf{B},\xi))\ dt\right), \tag{28}$$

where N is the total number of simulated particles.

Since we simulate very large N (around $1e6$ particles), the expected value might be very low even if we multiply by energies (which are indeed very high). Keep in mind that we also hope $D(\mathbf{B},\xi)$ to be as low as possible. So, to avoid issues with the objective function being too flat and the BO struggling to find a minimum, we take the log of our simulation output. At the end, our minimization problem becomes:

$$\min_{\theta\in\Theta}\log\left(\mathcal{J}(\theta)+\varepsilon\right). \tag{29}$$

The study of robustness of BO under noise will come in the next chapters, but we already have some guarantees in literature, like in [12], where Gardner et al. provide bounds for local optimization in high dimension under noisy conditions.

5 The field parametrization problem

5.1 The coil method

We need now to define the parameters space Θ that controls the magnetic field. The only requirements for BO to work are:

- Θ must be compact,
- $\mathbf{B}_\theta : \Theta \to \mathcal{F} \in C^0(\Theta,\mathcal{F})$.

The most obvious choice for a real world physics application is to parametrize an array of single coils. This ensures that the resulting optimized field is physically reproducible and that it respects the PDE constraint $\nabla\cdot\mathbf{B} = 0$, which would be hard to enforce in the optimization algorithm. For each coil we have the following parameters:

1. $r \geq 0$ is the distance of the coil from origin;
2. $\psi \in [0,\pi]$ is azimuthal angle of coil center vector;
3. $\phi \in [0,2\pi]$ is polar angle of coil center vector;
4. $\alpha \in [0,\pi]$ is azimuthal angle of coil normal versor;
5. $\beta \in [0,2\pi]$ is polar angle of coil normal versor;

Also, we have a shared parameter $R \geq 0$ which is the radius of each coil. We can clearly see that if we choose finite bounds for r and R, compactness of Θ is ensured. The field components in the frame of reference of the coil come from the following integrals, obtained using Biot-Savart law:

$$B(x) = \frac{\mu_0 IR}{4\pi}\int_0^{2\pi}\frac{z\cos(\theta)}{(x^2+y^2+z^2+R^2-2Rx\cos(\theta)-2Ry\sin(\theta))^{3/2}}d\theta\ ;$$
$$B(y) = \frac{\mu_0 IR}{4\pi}\int_0^{2\pi}\frac{z\sin(\theta)}{(x^2+y^2+z^2+R^2-2Rx\cos(\theta)-2Ry\sin(\theta))^{3/2}}d\theta\ ;$$
$$B(z) = \frac{\mu_0 IR}{4\pi}\int_0^{2\pi}\frac{R-x\cos(\theta)-y\sin(\theta)}{(x^2+y^2+z^2+R^2-2Rx\cos(\theta)-2Ry\sin(\theta))^{3/2}}d\theta\ .$$

It comes as a consequence that the field is continuous everywhere except on the coil itself, where the magnitude explodes to infinity with speed $\mathcal{O}\left(\frac{1}{x^2}\right)$. This is actually not a problem for the simulation, since the probability of a particle hitting exactly the discontinuity point is 0, but it will require to weaken the assumption made in Proposition 2.

The other drawback is that every single coil requires five parameters, thus limiting the array to a maximum size of ~ 10, so that the optimization problem has ~ 50 dimensions, which should be manageable by BO, according to existing literature.

Some techniques have been proposed for dimensionality reduction in BO, like random embeddings in [15], but it comes at the price of losing the physical meaning of the original parameters, thus rendering it useless for this parametrization.

6 Code implementation

6.1 The Python optimizer

6.1.1 The Sobol initial sampling method

6.1.2 The BoTorch library functions

6.2 The C++ Monte Carlo simulation

6.2.1 Sampling from data with MC

6.2.2 The SAM algorithm

6.2.3 The Boris algorithm

7 Results and Conclusions

8 Remaining questions

There are still a few more question to address for this application of BO:

1. What are the guarantees of convergence under noisy conditions (crucial if the Monte Carlo simulation has high variance)?
2. How do this problem's convergence rate and regret bound scale with dimension?
3. What is the most useful parametrization, both from a mathematical and a physical perspective?

9 Code work

The next steps will be:

1. Verify if SAM integrator is faster than Biot Savart (DONE),
2. Implement sampling from Oltaris data for accurate particle simulation (DONE),
3. Experiment with different acquisition functions (current one is Expected Improvement) (TRIED KNOWLEDGE GRADIENT),
4. Change the kernel to accomodate periodic parameters instead of relying on feature map.
5. Need to implement interchangeability of coils.
6. Push the dimensionality of the problem to the limits of BO (WORKING): I need to implement either Trust Region algorithm to optimize locally or Random Embeddings to reduce dimensionality. Maybe try SAASBO.
7. Experiment with different parametrizations.
8. Reduce variance of Monte Carlo simulation, currently the highest number of particles that is doable is 28e4, so i need to reduce the variance. The easiest path is to use quasi Monte Carlo methods, like Sobol sampling and recycle the same starting points across different optimizations. If there will be time, i will implement a backward Monte Carlo (Adjoint MC) to do importance sampling of interesting trajectories.

Additional references include: [13], [16].

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